

HisabDo AI — Day 13

AI Forecast Explanation & Insights — Aarti

1. Module Objective

The purpose of this module is to convert verified cash-flow forecast results from the backend/forecasting engine into clear, natural-language financial insights. The AI explanation layer must explain verified values and must never calculate, modify, or invent financial amounts.

2. Aarti — Exact Task Scope

1. Define the AI explanation structure for forecast results.
2. Use only verified forecast data from the backend/forecasting engine.
3. Generate natural-language forecast insights.
4. Explain expected income and expenses.
5. Explain expected savings.
6. Explain expected closing balance.
7. Explain positive or negative cash flow trends.
8. Generate actionable observations from verified forecast results.
9. Clearly communicate forecast uncertainty or limitations where applicable.
10. Ensure AI does not invent financial amounts or predictions.
11. Prepare sample AI forecast responses for testing.
12. Handle missing or insufficient data safely.

3. Correct AI Architecture

HisabDo Database → Forecasting Engine → Verified Forecast Result → AI Explanation Layer → User

The forecasting engine is the source of truth. The AI/LLM is only an explanation layer. All numeric forecast values must already exist in the verified backend result before they are sent to the AI.

4. AI Explanation Structure

1. Forecast Summary: Briefly explain what the forecast indicates for the next period.

2. Expected Income: Explain the verified expected income and its context.
3. Expected Expenses: Explain the verified expected expense amount and the spending outlook.
4. Expected Savings: Explain expected savings using the verified forecast result.
5. Expected Closing Balance: Explain closing balance only when a trusted opening balance and forecast closing balance are available.
6. Cash Flow Trend: Explain whether the verified forecast indicates positive or negative cash flow.
7. Actionable Observation: Give a practical suggestion based only on verified forecast results.
8. Uncertainty & Limitations: Clearly state that forecasts are estimates and mention data limitations when provided by the backend.

5. Verified Backend Input

Recommended input object:

- ExpectedIncome
- ExpectedExpense
- ExpectedSaving
- OpeningBalance (optional)
- ExpectedClosingBalance (optional)
- CashFlowDirection
- Confidence
- HistoricalMonthsUsed
- Limitations

The AI should receive the verified values exactly as produced by the forecasting engine. If a value is missing, the AI must not estimate or guess it.

6. Explanation Rules

- Use ExpectedIncome exactly as provided by the backend.
- Use ExpectedExpense exactly as provided by the backend.
- Use ExpectedSaving exactly as provided by the backend.
- Use ExpectedClosingBalance only when it is supplied and marked valid.
- Explain positive cash flow when expected saving/cash flow is positive.
- Explain negative cash flow when expected saving/cash flow is negative.
- Mention confidence and limitations when available.
- Do not create a new forecast amount.
- Do not change rounding or percentages supplied by the backend.
- Do not claim certainty about future income or expenses.

7. AI No-Invention Rules — Mandatory

- Never invent income, expense, saving, or closing-balance amounts.
- Never change a backend forecast value.
- Never calculate an alternative forecast inside the LLM.
- Never invent a historical period or number of months.
- Never claim a forecast is guaranteed.
- Never fill missing financial values with assumptions unless the backend explicitly provides them.
- Never claim fraud, theft, or unauthorized activity from a forecast alone.
- If insufficient data is reported, clearly explain that the forecast is limited or unavailable.

8. Recommended AI Prompt

You are the HisabDo AI financial explanation assistant. Explain only the verified forecast data provided by the backend. Do not calculate new financial values and do not invent any amount, prediction, trend, or historical fact. If a value is missing, say that it is not available. Clearly communicate that forecasts are estimates and mention the supplied confidence or limitations. Keep the response simple, helpful, and actionable.

9. Sample AI Forecast Responses

Normal forecast

Based on your recent financial pattern, your expected income for the next month is PKR 60,000 and your expected expenses are PKR 42,000. Your expected savings are PKR 18,000. This indicates a positive cash flow outlook. The forecast is based on available historical data and may change if your income or spending behavior changes.

Negative cash flow

Your expected expenses for the next month are PKR 55,000 compared with expected income of PKR 45,000. This results in expected savings of -PKR 10,000, indicating a negative cash flow outlook. Consider reviewing non-essential expenses. This is an estimate based on available historical data and may change with your actual spending or income.

Closing balance available

Your forecast shows expected income of PKR 70,000 and expected expenses of PKR 50,000, giving expected savings of PKR 20,000. With the verified opening balance available, your expected closing balance is PKR 95,000. This forecast is an estimate and may change as your financial activity changes.

Limited history

A forecast is available, but it is based on limited historical data. Treat the expected income and expense values as an estimate rather than a certain prediction. The forecast may change as more financial history becomes available.

Insufficient data

There is not enough verified historical financial data to produce a reliable forecast at this time. No financial amount has been estimated. Add more income and expense history so HisabDo can generate a more useful forecast.

10. Missing Data Safety

The explanation service must safely handle null or unavailable forecast values. For example, if ExpectedClosingBalance is unavailable, the response should explain income, expenses, and savings without inventing a closing balance. If the complete forecast is unavailable because of insufficient data, the service should return a safe limitation message.

11. .NET Implementation Boundary

The C# service below prepares a deterministic, verified explanation context. An LLM provider (OpenAI, Gemini, Ollama, or another approved provider) can later receive this context. The provider must remain an explanation layer; numeric source-of-truth calculations stay in the forecasting engine.

12. Testing Scenarios

- Normal positive cash flow forecast.
- Negative cash flow forecast.
- Expected saving is positive.
- Expected saving is negative.
- Expected closing balance available.
- Opening balance missing.
- Expected closing balance missing.
- Low confidence / limited historical data.
- Insufficient historical data.
- Null financial values.
- AI prompt contains only verified backend values.
- AI response does not invent or alter financial amounts.

13. Aarti Deliverables

- Final forecast explanation structure.
- Verified forecast input contract.
- AI prompt and no-invention rules.
- Natural-language forecast insight logic.
- Sample forecast responses.
- Missing-data and limitation handling.
- Testing scenarios and expected outputs.

- Handoff specification for Laiba/backend integration.

14. Team Lead Acceptance Checklist

- Forecast explanation structure is finalized.
- Only verified backend forecast data is used.
- Expected income and expenses are explained.
- Expected savings are explained.
- Closing balance is explained only when valid.
- Positive/negative cash flow is explained.
- Actionable observations are based on verified results.
- Forecast uncertainty and limitations are communicated.
- AI cannot invent or alter financial values.
- Missing and insufficient data are handled safely.
- Sample responses pass validation.

15. Final Definition

The Day 13 AI Forecast Explanation Layer converts verified cash-flow forecast results into simple, understandable, and actionable natural-language insights. The backend/forecasting engine remains the source of truth for every financial amount and prediction. AI explains the verified results, communicates uncertainty, and never invents financial values.